Experiment

① 8 bit - Addition & Subtraction

A = 9C H

A = 1001 1100

B = 7A H

B = 0111 1010

Addition: A + B => (9C + 7A) H.

A + B => 1001 1100

0111 1010 (+)

10001 0110

A + B = 16 H

Cy = 1.

Subtraction: A - B => (9C - 7A) H.

A - B => 1001 1100

0111 1010

00010 010

8-bit Multiplication &

B) 16-bit Addition & Subtraction

X = 1234 H

Y = 5678 H

X + Y = 1234 + 5678

LSB = (34 + 78) H

LSB ⇒ 1000 / 0

100 / 110

LSB ⇒ 00110100

+ 01111000

10101100

MSB = 12H + 56H

MSB ⇒

MSB ⇒ 0 0010010

0 1010110

0 1101000

MSB ⇒ 68 H

MSB + LSB ⇒ 0110100010101100

X + Y ⇒ 68 AC H

Subtraction.

X - Y ⇒ 1234 H - 5678 H

⇒ -4444 H

y into 2's complement.

y = 01010110 01111000

1's complement ⇒ 1010 1001 1000 0111

Add 1 ⇒ 1010 1001 1000 1000

X - Y ⇒ 0001 0010 0011 0100

1101 0110 1100 1000

1011 1011 1011 1100

X - Y ⇒ BBBCH

8-bit Multiplication & Division

Multiplication : 12H x 05 H.

Hexadecimal to decimal.

12 H => (1x16¹) + (2x16⁰)

= 16 + 2.

= 18

05 => (0x16¹) + (5x16⁰)

=> 5

18 x 5 => 90

Decimal to Hexadecimal.

90 ÷ 16 => 5 (10)

=> 5A H.

Division: 64H ÷ 05 H.

64 H => (6x16¹) + (4x16⁰)

=> 96 + 4

=> 100

05 H = 5

100 ÷ 5 => 20

20 ÷ 16 => 14 H.